

Recent advances on image edge detection: A comprehensive review^{*,**}

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ABSTRACT

Edge detection is one of the most important and fundamental problems in the field of computer vision and image processing. Edge contours extracted from images are widely used as critical cues for various image understanding tasks such as image segmentation, object detection, image retrieval, and corner detection. The purpose of this paper is to review the latest developments on image edge detection. Firstly, the definition and properties of edges are introduced. Secondly, the existing edge detection methods are classified and introduced in detail. Thirdly, the existing widely used datasets and evaluation criteria for edge detection methods are summarized. Finally, future research directions for edge detection are elaborated.

1. Introduction

Edge detection is one of the most important tasks in the field of computer vision and image processing, and edge provides important information for many subsequent visual tasks, such as image recognition [116, 121, 41, 163, 40, 140], image segmentation [169, 9, 167], image retrieval [124, 59, 93, 137, 88], face recognition [192, 24, 89], corner detection [189, 184, 187], road detection [37], medical imaging [136], and target tracking [27, 87, 177, 120]. Each of these visual tasks requires extracting object boundaries or perceiving obvious edges from the original image. Therefore, the stability detection of image edge details in order to efficiently perform the actual task.

The existing surveys about image edge detection have been summarized in Table 1. It can be seen from Table 1 that the existing edge detection surveys [103, 91] specifically focus on hand-crafted edge detection techniques such as active contour based methods [26, 20], multiscale based methods [28, 61, 132], and anisotropic diffusion based methods [62, 23, 117, 155]. Our investigation also indicates that some of the methods mentioned in existing surveys are outdated and there lacks of a survey on latest developments on image edge detection, especially the review on deep learning [172, 15] based edge detection methods. This paper intends to fill this gap, and the latest developments on image edge detection methods are introduced and summarized. The main contributions in this work comprise three aspects. Firstly, the recent edge detection methods are classified and introduced in detail. Secondly, the existing widely used datasets and evaluation criteria for edge detection methods are reviewed. Thirdly, future research directions for edge detection are elaborated.

The remainder of this paper is arranged as follows. In Section 2, related background and main challenges on image edge detection are introduced. In Section 3, the existing edge detection methods are classified and introduced in detail. In Section 4, the existing widely used datasets and evaluation criteria for edge detection methods are introduced. In Section 5, the future research directions for edge detection are described, and conclusions are given in Section 6.

2. Generic edge detection

In this section, a brief background on the definition and properties of an edge contour is presented. And then the main challenges on image edge detection are introduced.

2.1. Background

In [157], an edge contour is defined as a collection of points with sudden brightness changes on an image. These sudden changes in brightness may be due to color changes, grayscale changes, or texture changes in the image. It is indicated in [149] that edges exist between regions or between the objects and background in the image. Due to the difference in imaging conditions, illuminations, scales, noise, viewpoints, and imaging qualities, the description and prediction of edge points will also be different. Our investigation shows that the edge detection criteria proposed by Canny [21] is comprehensive, and the criteria are unique response to one edge, good edge localization, and insensitivity to noise.

2.2. Main challenges

The ideal goal of image edge detection is to develop a general-purpose method which has the ability to achieve high precision and high efficiency at the same time as shown in Fig. 1.

High detection accuracy means that the designed edge detector should have the ability to detect real edge contours as accurately as possible, detect as few false edges as

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Table 1

Summary of related image edge detection surveys.

No.	Survey title	References	Year	Venue	Content
1	A survey of edge detection techniques	Davis [30]	1975	Computer Graphics and Image Processing	A survey of linear and prior knowledge.
2	Theory of edge detection	Marr et al. [110]	1980	Biological Sciences	A survey of second edge order detector at a given scale.
3	Theory of edge detection	Peli et al. [126]	1982	Computer Graphics and Image Processing	A survey of the performance of different noises.
4	On edge detection	Wang et al. [157]	1986	Computer Vision and Pattern Recognition	A survey of differential operators.
5	Localization and noise in edge detection	De Micheli et al. [31]	1989	Computer Vision and Pattern Recognition	A survey of the positioning accuracy sensitivity under noise.
6	Edge detection techniques-An overview	Ziou et al. [193]	1998	Pattern Recognition and Image Analysis	The mutual influence between edges and detectors.
7	Gaussian-based edge-detection methods: A survey	Basu [13]	2002	Transactions on Systems, Man, and Cybernetics, Part C	A survey of linear and nonlinear Gaussian filters and their characteristics.
8	A classified and comparative study of edge detection algorithms	Sharifi et al. [146]	2002	International Conference on Information Technology	A survey of five kinds of edge detectors.
9	Edge detection techniques: Evaluations and comparisons	Nadernejad et al. [118]	2008	Applied Mathematical Sciences	Performance of several edge detectors are introduced.
10	Performance evaluation of edge detection techniques for images in spatial domain	Juneja et al. [79]	2009	International Journal of Computer Theory and Engineering	A survey of detectors' performance by statistical evaluation criteria.
11	A survey on edge detection methods	Oskoei et al. [122]	2010	University of Essex	A survey of edge detectors before 2010.
12	Edge and line oriented contour detection: State of the art	Papari et al. [123]	2011	Image and Vision Computing	A survey of quantitative evaluation of contour detection.
13	A comparison of various edge detection techniques used in image processing	Shrivakshan et al. [147]	2012	International Journal of Computer Science Issues	A survey of common filter technology and application.
14	A survey on various edge detector techniques	Bhardwaj et al. [16]	2012	Procedia Technology	A survey of valuation various differential detectors.
15	Quantitative error measures for edge detection	Lopez-Molina et al. [103]	2013	Pattern Recognition	A survey of the quantitative evaluation algorithm under different image sets.
16	A survey of recent advances in visual feature detection	Li et al. [91]	2015	Neurocomputing	A survey of evaluation multiple visual features.

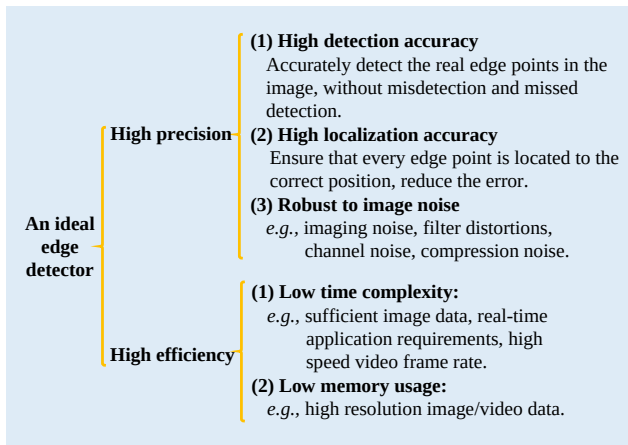


Figure 1: Main challenges for image edge detection.

possible, and have good edge localization capabilities. The reason is that edge contours based computer vision or image processing tasks utilize edge contour for establishing the relationship between different feature regions such as those for image segmentation and image recognition. Therefore, high-precision edge detectors should have high robustness under different imaging conditions (e.g., different camera poses and different camera imaging resolutions) and robust to different types of noise. Furthermore, a high precision/quality edge detection method should have the ability to extract edge contours in real-time with acceptable memory and storage demands. The reason is that real-time processing is an indispensable requirement in many edge contour based computer vision tasks such as image segmentation [169, 9].

3. The classification of image edge detection methods

In this section, the existing edge detection methods are introduced and summarized. Our investigation indicates that the existing edge detection methods can be divided into two categories: hand-crafted based methods [110, 188, 76, 85, 164] and machine learning based methods [178, 174, 90, 172, 73]. The representative image edge detection methods are shown in Fig. 2. We will review the two main streams in the edge detection literature as follows.

3.1. Hand-crafted based edge detection

The existing hand-crafted based edge detection methods include differentiation based [154, 188, 164], anisotropic diffusion based [104, 127], active contour based [81, 12, 77, 143, 46], fuzzy logic based [83, 162, 161, 3, 6, 35], spatial-frequency based [86, 43, 4, 80, 10], statistics based [85, 139, 5, 171, 128, 101], and sub-pixel interpolation based [58, 70, 17, 29, 22, 182] methods. We will review the existing edge detection techniques in the hand-crafted based edge detection literature as follows.

Differentiation-based methods. The existing differentiation-based edge detection methods mainly utilize first-order or

second-order derivatives of the light intensity of an image with single or multiple scales for detecting edge contours from images.

Early monoscale methods (e.g., Sobel method [152]) employ simple kernels or operators for obtaining the edge strength maps. Although efficient, these early methods are sensitive to noise [188], and therefore, some monoscale approaches employ kernels with variable sizes. In [110], the Laplacian of Gaussian kernels are employed for yielding the edge response. By setting different kernel sizes, one can apply the Laplacian of Gaussian kernels to either thin or wide edges. Canny detector [21] is a more advanced method and has gained the most popularity for edge detection. This method, which employs the first-order derivative of isotropic Gaussian kernel, has been considered as the theoretically optimal operator for detecting isolated step edges corrupted by additive white Gaussian noise. However, the scale choice in the Canny method is still an issue. To overcome this issue, many edge detectors [36, 74, 176, 76] have been developed to improve the Canny method. Zhang et al. [188] utilize the anisotropic Gaussian filters [185, 186] to obtain anisotropic intensity variation information for detecting edge contours from images. It is proved that this type of anisotropic kernel has the ability to obtain a good noise-robustness while maintaining a good edge detection resolution. Nevertheless, anisotropy kernels can incur a so-called anisotropy stretch effect, yielding spurious results in the vicinity of blobs and corners [188]. Using a similar strategy, anisotropic morphological directional derivatives are utilized to suppress the impulse-noise for edge detection [148].

Edges in natural images usually extend over different scales [45], and as a result, those monoscale detectors degrade when detecting edges with heterogeneous widths [45, 157]. Inspired by the mechanism of the human visual system [11], multiscale edge detection has gained increasing attention [105]. One popular scheme is to fuse multiple edge strength maps obtained at different scales. For example, the method proposed in [149] yields the fused edge strength by computing the product of the gradients obtained at two different scales. Shui and Zhang [149] designed a method based on both isotropic and anisotropic Gaussian kernels, computing the geometric mean of two gradient maps as the edge strength. However, these methods only consider gradient responses at two scales, showing limitations in handling multiscale edges. In [45], the authors proposed a scale estimation method based on the second-order derivative of the image intensity, but this method requires prior knowledge of sensor noise. Based on the scale-space theory, Lindeberg [95] developed an automatic scale selection approach for edge detection based on the scale-normalized Gaussian kernels. To extend the anisotropic edge detection methods into image scale-space, Wang et al. [164] proposed the scale-normalized anisotropic Gaussian kernels. These kernels yield the maximum response on the matched orientation and at the matched scale, and therefore, this method utilizes maximum response in scale-space to identify edge scales.

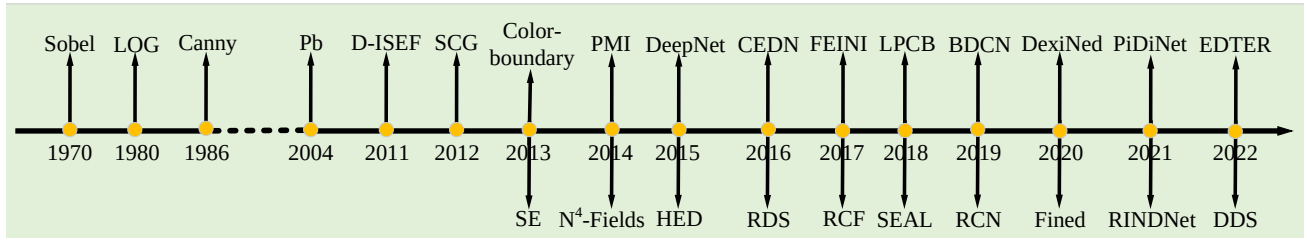


Figure 2: Representative image edge detection algorithms. Starting from Sobel algorithm [152], differentiation based edge detection algorithms became extraordinarily popular. In 1980, Marr and Hildreth [110] proposed a Laplace-of-Gaussian method for detecting edge contours which represents the beginning of second-order derivative based edge detection algorithm. In 1986, Canny [21] proposed a first-order derivative based edge detection algorithm which gained great popularity. Inspired by the Canny method, first-order directional derivative based edge detection methods (e.g., D-ISEF [112], color-boundary [180], and FDAG [188]) have been presented for detecting edge contours from images. In 2004, Martin et al. proposed an edge probability (Pb) method [111] for predicting edges based on the characteristics of information theory. In 2015, the HED method [174] is proposed which utilizes an edge-to-end supervised model for learning edges. Many algorithms (e.g., CEDN [178], RCF [99], and LPCB [34]) extended the HED method based on a given backbone network (e.g., VGG and ResNet) and achieved good edge detection accuracy. In 2019, BDCN [66] is proposed to extract edges at dramatically different scales. Recently, many light network architectures (e.g., Fined [170], EDTER [135], and PiDiNet [153]) are specifically designed for detecting edge contours from images which do not require a large number of datasets and pre-trained models.

In addition, an adaptive anisotropy factor is proposed for improving the noise robustness of the small-scale edges and alleviating the anisotropy stretching effect simultaneously.

Statistical methods. Although differentiation-based methods have good efficiency and interpretability, they can hardly suppress undesired textures. Meanwhile, differentiation-based methods cannot be used to identify edge structures defined by textures. Moreover, the key parameters in such edge detection methods should be carefully chosen by the users. To overcome the aforementioned limitations, statistical edge detection methods were proposed to analyze the local image pattern around each pixel. In [139], the edge thresholding procedure is performed using statistical principles. Local standardization of thresholds for each pixel is determined by the statistical variability of the gradient vector. Then, such a standardized statistics based on the gradient vector at each pixel is used to determine the eligibility of the pixel to be an edge pixel. Knoishi et al. [85] formulated edge detection as data-driven statistical inference. For any set of edge detection filters, the pre-segmented images are used to learn the probability distributions of filter responses conditioned on whether they are evaluated on or off an edge. Edge detection is subsequently formulated as a discrimination task specified by a likelihood ratio test on the filtering result. Paying attention to suppressing of the off-edges, this method improves the edge detection performance when the image contains a complex textural background.

Fuzzy-based methods. Since Russo [144] introduced fuzzy inference to efficiently extract edge features with strong noise robustness, the application of fuzzy theory in edge detection has attracted increasing attentions. Because the definition of edges in an image inherently has uncertainty, the use of fuzzy theory is suitable for detecting edge structures. Versaci and Morabito [162] proposed an edge detector based on the joint use of fuzzy divergence and fuzzy entropy minimization, solving the edge detection problems

for gray-level images characterized by uncertainty, imprecision, or both. Its performance is similar to the Canny detection, but this method has a lower computational complexity. Melin et al. [114] proposed an edge detector based on the morphological gradient technique and generalized the type-2 fuzzy logic. In the edge detection procedure, they employed the theory of alpha planes to implement the generalized type-2 fuzzy logic. A good edge detection performance was obtained during the evaluation test in terms of the merit of Pratt measure.

Spatial-frequency based methods. Random noise is ubiquitous in digital images, and therefore, the edge and noise in the spatial domain show large fluctuations in gray scale. In the frequency domain, they are usually reflected as the same main frequency components, which makes real edge detection a difficult task [108]. To tackle this problem in the frequency domain, edge detection methods based on multi-scale spatial-frequency analysis were developed. Since wavelet analysis is a powerful tool and is closely related to multi-scale analysis, Mallat et al. [107] proposed a local maximum modulus edge detection method based on wavelet transform multi-scale analysis. Wavelet transform has good time-frequency local conversion and multi-scale analysis capabilities, so it is more practical and accurate than other types of edge detection strategies. Furthermore, a shearlet transform method based on multi-scale direction is proposed in [181], which has a stronger ability on local distribution discontinuity. The shearlet algorithm is very effective in detecting edge position and direction, which is widely considered superior to other standard methods based on wavelet and others.

Sub-pixel localization methods. Precisely, estimating edge characteristics in images, such as the position, orientation, curvature, and local contrast, based on the gradient vector in each pixel usually has a certain error. In order to cope with the problem of positioning accuracy in edge

detection, sub-pixel interpolation has also been introduced into edge detection [58]. Adopting such strategies, Trujillo-Pino et al. [158] proposed an edge detector based on an edge and acquisition model derived from the partial area effect, which does not assume continuity in the image values but assumes that the edge could be approximated by straight lines in the neighborhood of each edge pixel, and then adapted to second order curves. In case of a low level of additive noise, the expressions of the method are rewritten to be applied to a smoothed version of the original image. When the noise is very high, an iterative method for enhancing the image quality is employed to maintain the precision of the obtained features. Seo [145] proposed a sub-pixel edge localization method based on the adaptive weighting of gradients. The method firstly finds the optimal power factors to be used by the adaptive weighting of gradients through an error propagation scheme. Subsequently, the edge width is directly related to the optimal powers. Also, the method uses the squared weighting of gradients, which applies a power of two to the gradients of an edge profile to calculate the edge location with sub-pixel accuracy. Comparatively, this method has a lower computational cost but higher precision.

Anisotropic diffusion based methods. Many edge detection methods, especially the differentiation-based methods designed based on Gaussian kernels, will inevitably incur a Gaussian blur effect. Different from anisotropic Gaussian filtering, anisotropic diffusion filtering is mainly used to smooth an image, since it preserves the edges of the image while smoothing the image [18]. Perona and Malik [127] presented a class of algorithms used to realize a diffusion process. The diffusion coefficient is chosen to vary spatially in such a way as to encourage intraregional smoothing rather than interregional smoothing. As the region boundaries in the approach remain sharp, a high-quality edge detector that successfully exploits global information can yield accurate edge detection results. Filters that can yield similar image processing results have also been found in bilateral filtering [156], adaptive bilateral filtering [55], and guided image filtering [67]. Since anisotropic diffusion filtering is usually implemented iteratively, a stop criterion is proposed in [159], which takes into account both image de-noising and accurate edge feature retention for detecting edge contours from images.

Active contour based methods. An outline can also be considered as an edge in some scenarios [81]. However, lacking global visual information, local edge detectors can hardly delineate discontinuous outlines. Therefore, active contour based methods are also proposed for edge detection. Such a method uses a function that approximates the original image continuously, and then uses differential derivation to obtain the gradient information to represent edge points. Subsequently, contour representation is used to fit the edge points [143]. For example, Bastan et al. [12] proposed to use the continuity and smoothing information of local pixels to fill the missing pixels and connect the discontinuous edges. In a representative application, the active contour method is used to detect edge contours of the coastline [97].

The continuous coarse coastline is obtained by the region-based method based on the low-resolution image obtained by multi-view processing. Then a fast snake-like active contour model based on edge characteristics is used to refine the coarse detection results.

3.2. Machine learning based methods

With the development of texture feature description techniques [106, 166, 42, 71], Martin et al. [111] employed image local cues (i.e., brightness gradient, texture gradient, and color gradient information) for presenting a probabilistic boundary (Pb) based edge detection method. In [111], logistic regression is utilized to combine the χ -distance of the image local cues (i.e., brightness gradient, texture gradient, and color gradient information) and learn a discriminative model for detecting edge contours from images. Ren [142] extended the Pb method [111] and presented a multi-scale probabilistic boundary (MsPb) based edge detection method which utilizes local boundary cues, i.e., contrast, localization, and relative contrast, and combines strengths from large scale and small scale for detecting edge contours with robustness to noise and detail-preserving. Arbelaez et al. [7] extended the Pb method [111] and presented a global probabilistic boundary (g-Pb) method which uses multi-scale and spectral clustering for obtaining local cues and detecting edge contours from images. In [125], texture based Pb (tPb) is proposed which uses the average of texture variations in randomly placed pairs of windows for estimating discriminative edge responses. It is indicated in [179] that the aforementioned edge detection approaches [111, 142, 7, 125] lift edge detection technique to a higher abstract level, but they lack object-level knowledge and still fall below human perception.

In recent years, deep learning techniques have made unprecedented progress in the field of computer vision and image recognition which are also widely used for image edge detection. Depending on whether the datasets have intuitive label features, the existing deep learning based edge detection methods can be divided into supervised learning based (i.e., with label features) image edge detection methods [2, 51, 98, 100, 183] and unsupervised learning based (i.e., without label features) image edge detection methods [141, 96, 52, 78, 56, 53].

Supervised learning based image edge detection.

The current image processing tasks are largely carried out via supervised learning. In this case, each sample has a corresponding label, and a deep neural network (e.g., ResNet [68]) is employed to learn the characteristics corresponding to each label for performing different visual tasks. A schematic of supervised learning based image edge detection is shown in Fig. 3.

In [37], an extended version of the probabilistic boosting tree classification method [160] is proposed which utilizes a large number of features with multiple scales for learning a discriminative model for detecting edges and object boundaries from images. Based on an ant colony algorithm [39], Rahebi et al. [138] used an artificial neural network [47]

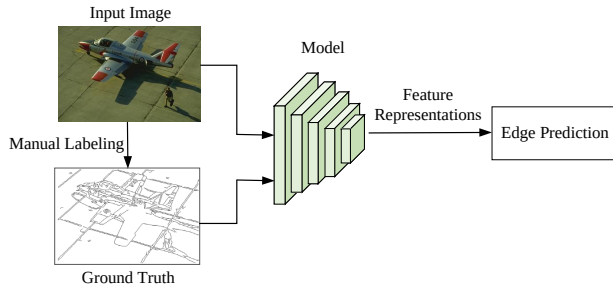


Figure 3: A schematic of supervised learning based image edge detection.

with supervised learning and momentum for improving biomedical image edge detection. In [92], small chunks of human-generated contours are clustered into sketch marker classes [44], and then edge contours are effectively detected using a random forest classifier. In [38], the problem of image edge detection is formulated as predicting image local segmentation masks based on random decision forest [57]. In [?], neural network nearest neighbor fields is designed for extracting edges which is the first deep convolutional neural networks (CNNs) based edge detection method that the nearest neighbour technique is utilized to match the extracted features from image patches to detect edge contours from images. In [174], a holistically-nested edge detection (HED) method is proposed which makes full use of convolutional neural network [15] and a deep supervised network [151] for image-to-image prediction. Furthermore, guided by deep supervision network [151], the HED method has the ability to automatically learn rich hierarchical representations which can be effectively utilized to address challenging ambiguities in edge and object boundary detection. In [66], VGG16 [151] is pre-trained on the ImageNet [32] as a backbone and a bi-directional cascade network (BDCN) is designed for extracting edges. A scale enhancement module in BDCN is specifically designed which has the ability to utilize dilated convolution to generate multi-scale features and detect edge contours at dramatically different scales. In [99], to effectively utilize hierarchical CNN features for classifying pixels to edge or non-edge sets, the CNN features of all the convolution layers are employed to perform the pixelwise prediction in an image-to-image fashion for extracting edge contours from images.

Unsupervised learning based image edge detection. It is indicated in [165] that the cost of labeling high-quality dataset increases exponentially with the number of targets and the difficulty of identification. Unsupervised learning based image edge detection does not need manual marking edge features for training, and it can estimate edge contours from the basic knowledge of the image itself as shown in Fig. 4.

It is indicated in [173] that the accuracy of edge contour detection can be significantly improved by obtaining sparse code gradients (SCG) [25], and the classification of sparse code gradients based on the multi-scale pooling and the

power transformation have the ability to effectively learn how to measure local contrast and find edge contours before using linear support vector machines (SVM) [69] for image edge classification. In [73], based on the basic principle that pixels belonging to the same target show higher statistical correlation than pixels belonging to different targets, pointwise mutual information architecture [50] is employed for extracting edge contours from images. In [179], a fully convolutional encoder-decoder network based on full convolution network [102] and unpooling [119] is developed for detecting object contours from images. The model is trained on the PASCAL VOC dataset [49] which has the ability to generalize well to unseen object classes from the same super-categories. In [90], a novel edge detection method is proposed that edge feature information is learned from video frames which indicates that the accuracy of motion estimation [14] and edge detection can be greatly improved by DeepMatching [113] for obtaining semi-dense matches between video frames. In [172], an encoder-decoder architecture for unsupervised semantic segmentation has been proposed for detecting edge contours, in which the encoder encodes an input image into a multi-class segmentation map that is then decoded to produce the original input. The segmentation map is constrained by a soft cut loss and post-processed by conditional random fields and hierarchical merging of areas.

4. Datasets and performance evaluation

In this section, the widely used datasets and performance evaluation criteria for image edge detection are introduced and summarized.

4.1. Popular datasets for image edge detection

It is well-known that datasets play a key role for image edge detection. On the one hand, datasets are utilized to evaluate the performances of image edge detection methods. On the other hand, datasets are utilized to greatly promote the development of the field of image edge detection. The existing popular datasets (e.g., the Berkely segmentation dataset (BSDS500) [7] and multicue dataset for boundary detection (MDBD) [115]) for image edge detection and their corresponding download links are summarized in Table 2.

In addition to important dataset as the basis for evaluation, a good edge detector should have good performance in detection accuracy and detection efficiency in order to realize practical applications for image processing tasks. It is worth to note that most computer vision or image processing tasks, e.g., image segmentation and image recognition, utilize edge detection for establishing the relationship between different feature regions. Therefore, high-precision edge detectors should have high robustness under different imaging conditions (e.g., different camera poses, different light sources, and different camera imaging resolutions). Therefore, the robustness of edge detectors should be measured by evaluating their detection accuracy, positioning error, and robustness to noise. In addition, the processing speed and memory consumption should also meet real-time

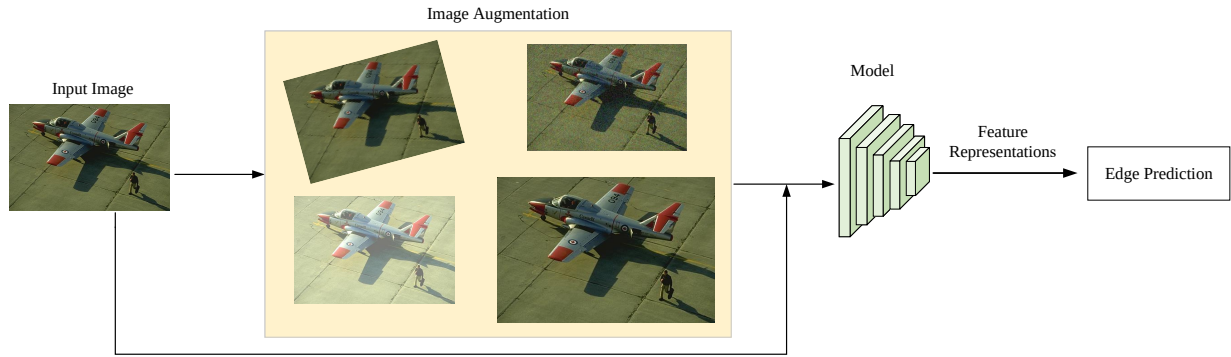


Figure 4: A schematic of an unsupervised learning based image edge detection.

Table 2

Popular datasets for performance evaluation on image edge detection.

Dataset name	Total images	Categories	Image size	Started year	Highlights	Download
Bowyer [19]	100	2	1280×720	2001	It consists of short binocular video sequences of natural scenes. And each scene has 5 boundary labels and 6 edge labels.	Bowyer
RuG40 [60]	40	2	512×512	2003	Each image represents a natural image, and its corresponding ground truth is also provided.	RuG40
BSDS [7]	500	3	481×321	2011	Each image represents a natural image, and its corresponding ground truth is also provided.	BSDS
PASCAL [48]	11,355	20	500×375	2012	It mainly contains images of people, animals, vehicles, and indoor scenes.	PASCAL
NYUD [150]	1449	3	560×425	2013	It consists of 381 training images and 654 test images.	NYUD
COCO [94]	300,000+	80	429×640	2014	It contains more than 1.5 million target object instances.	COCO
Multi-cue [115]	200	2	1280×720	2016	Each scene consists of left view and right view.	Multi-cue
MDBD [115]	28	2	512×512	2016	It contains artificially synthesized and naturally shot images.	MDBD

or acceptable storage range which must be considered in actual applications for different computer vision tasks.

4.2. Evaluation criteria

Our investigation indicates that the existing popular edge detection evaluation criteria include receiver operating characteristics (ROC) [19], figure of merit (FoM) [133], F-measure [111] at optimal dataset scale (ODS) and optimal image scale (OIS), average precision (AP), mean square error (MSE), signal-to-noise ratio (SNR), and time consumption. The aforementioned edge detection evaluation criteria can be divided into four categories: (1) the evaluation of detection accuracy on the ground truth; (2) positioning accuracy indicators on the ground truth; (3) robustness

of the detector under image affine transformation; and (4) execution time and memory consumption.

Detection capability and positioning accuracy are often used for performance evaluation on edge detection. It can quantitatively and intuitively use a given dataset to represent the score of the detection results, so that people can determine whether a real edge is detected and what the accuracy of edge point positioning is. It is worth to note that it is impossible to label a large number of dataset. To solve the aforementioned problem, scholars have developed the Figure of merit (FoM) [133] evaluation criteria which does not require ground truths.

Quantitative evaluation on precision error. Receiver operating characteristics (ROC) [19] is usually employed

for evaluating the performance of traditional edge detectors in noiseless and noisy color or gray images. It needs to identify the authenticity of detected pixels with the help of the ground truths (GTs) value of real labels. In the detected edge map, the detected pixels in the neighborhood of GT edge pixels are true positive (TP). False positives (FPs) consist of pixels detected in non-edge regions of GT, and pixels detected in unconcerned regions are ignored. Let n_{TP} and n_{FP} be the pixel numbers of true detection and false detection respectively. Let n_e and n_{ne} be the number of edge pixels and non-edge pixels in ground truth. The ROC mismatch rate (also known as false negative rate) and FP ratio are then calculated as follows

$$\begin{aligned}\gamma_{\text{unmatch}} &= 1 - \frac{n_{TP}}{n_e}, \\ \gamma_{FP} &= \frac{n_{FP}}{n_{ne}}.\end{aligned}\quad (1)$$

In Equation (1), an evaluation coordinate system is utilized which contains vertical axis γ_{FP} and horizontal axis γ_{unmatch} . Therefore, it can be concluded that the closer the ROC curve is to the lower left corner of the coordinate system, the higher the detection accuracy of the detector is, and the closer the detection result is to the ground truth. Parameter settings of an edge detector are first trained for attaining the best ROC curve on a single image and then tested on the set of images for obtaining the aggregate test ROC curve by averaging the aggregate ROC curves of all the images.

F-measure. Martin et al. [111] developed an evaluation method based on the F-measure on the BSDS dataset. Accuracy and recall rate are two indexes which are utilized for evaluating the detector performance in this evaluation criteria. Accuracy rate is used to measure the accuracy of the edge detection results. N_d is used to represent the number of all detected edge pixels, N_{Td} is the number of true detected edge pixels, N_{gt} is used to represent the number of the ground truth pixels. In this way, the accuracy is defined as

$$\text{Precision} = \frac{N_{Td}}{N_d}.\quad (2)$$

Recall is defined as the ratio of N_{Td} and N_{gt}

$$\text{Recall} = \frac{N_{Td}}{N_{gt}}.\quad (3)$$

Then the F-measure is defined as the harmonic mean of precision and recall

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}},\quad (4)$$

which represents the overall performance of an edge detector for an input image. It is worth to note that a threshold needs to be set to generate the final edge map for an edge probability map. There exists two ways for setting thresholds. One is the ODS technique which selects a fixed threshold for all

test images. Another way is the OIS technique which selects an optimal threshold for each image. In the precision-recall coordinate system, precision is used to denote the vertical axis and recall is used to represent the horizontal axis. And then a precision-recall evaluation coordinate system is constructed. The area under the precision-recall curve is used as an average accuracy measure of performance. In the precision recall coordinate system, the detector located at the upper right corner demonstrates that the detector has better edge detection performance.

Signal-to-noise (SNR). SNR is defined as

$$\text{SNR} = 10 \lg \left(\frac{(I_i(m_x, n_y))^2}{(I_d(m_x, n_y) - I_i(m_x, n_y))^2} \right),\quad (5)$$

where I_i represents the gray value of the ideal ground truth value at pixel (m_x, n_y) of an image, I_d represents the gray value of the detected edge map at pixel (m_x, n_y) of the image. In this way, the SNR is employed for evaluating the robustness of an edge detector for noise. In general, the larger SNR is, the lower the false detection rate and the better detection performance of a given edge detector.

Mean absolute error (MAE). MAE is used to represent the average distance between the detected edge pixels and the ground truth as given below

$$\text{MAE} = \frac{1}{m_x n_y} |I_d(m_x, n_y) - I_i(m_x, n_y)|,\quad (6)$$

where I_i denotes the gray value of the ideal ground truth value at pixel (m_x, n_y) of the image, and I_d represents the gray value of the detected edge map at the point of the image (m_x, n_y) . Thus it can be found that the larger the MAE is, the larger the error of the detected edges is.

Figure of merit (FoM). The evaluation criteria on FoM [133] does not need to use the ground truth for evaluating the performance of an edge detector. The score of FoM is defined as follows

$$\text{FoM} = \frac{1}{\max(n_e, n_d)} \sum_{k=1}^{n_d} \frac{1}{1 + \omega d(k)^2},\quad (7)$$

where n_e is the ideal number of edge points, n_d is the number of edge points actually detected by a given edge detector, the distance difference between the k^{th} detected edge point and the ideal edge point is denoted as $d(k)$, and ω is a constant. The ideal edge point is obtained by several detectors compared with a given edge detector for obtaining the edge map. The range of FoM is within [0-1]. 1 is the highest rate between the detected edge image and the ideal edge image.

The performance evaluation of state-of-the-art algorithms [21, 125, 8, 63, 92, 131, 130, 25, 129, 65] based on commonly used datasets, i.e., BSDS [7], NYUD [150], Multicue [115], and PASCAL [48], are performed and their corresponding results are summarized in Tables 3, 4, 5, and 6 respectively. It can be easily found from Tables 3, 4, 5, and 6 that the edge detection methods based on deep learning

techniques have achieved better evaluation results than the edge detection methods based on hand-crafted techniques. The reason is that the existing hand-crafted edge detection algorithms usually extract edge contours from images by calculating gradient information, but these methods can only obtain shallow edge structure information and do not have the ability to obtain rich multi-level hierarchical edge feature information. As it is indicated in [37], an edge cannot be accurately detected by low levels cues such as gradient. On the contrary, deep learning based edge detection methods have the ability to obtain rich multi-level hierarchical feature information, e.g., brightness, texture, color, and maybe contextual information, for detecting edge contours. It is worth to note that deep learning based edge detection methods have high computational complexity for obtaining better accuracy of edge detection results. To address the computational complexity of deep learning based edge detection, many lightweight neural network architectures have also been presented (e.g., EDTER [135], PiDiNet [153], and DSCD [33]).

5. Some future challenges on image edge detection

Despite the recent tremendous progress in the field of image edge detection, there are still open problems and some challenges.

- A more effective task-specific edge detector is necessary: Since the existing image edge detection methods have not considered how to effectively detect edge contours from different types of tasks or different types of images (e.g., from remote sensing imaging and optical imaging), it remains vital to develop different types of task-specific edge detection methods that have the ability to accurately detect edge contours in a task-specific manner.

- It is necessary to develop more advanced neural network frameworks for image edge detection: With the development of deep CNN architectures, the accuracies of image edge detection algorithms are becoming increasingly comparable. Nevertheless, there still exist much of works to be explored as illustrated in the following:

(1) Image edge detection under different imaging conditions. The existing image edge detection methods still cannot detect edge contours from the same scene under different imaging conditions well. Take a building scene [1] as an example, edge detection results from the HED method [174] under different illuminations are illustrated in Fig. 5. It can be found that some obvious edge contours in the window area (marked by ‘□’) cannot be detected under different illuminations as shown in Fig. 5(b) and (d).

(2) Adjacent image edge detection. The existing image edge detection methods still cannot accurately detect adjacent edge contours from an input image. Take a Chinese chessboard image [75] as an example, the edge detection results from the HED method [174] for this region are illustrated in Fig. 6(b). It can be found that some adjacent edge contours (marked by ‘○’) cannot be accurately detected by the HED method.

Table 3

Comparison of evaluation results of state-of-the-art algorithms on the BSDS500 dataset [7].

Method	ODS	OIS	AP	F-measure
Human	.803	.803		.803
Canny [21]	.600	.640	.580	.600
SLEDGE [125]	.672	.681	.652	.670
gPb_ucm [8]	.729	.755	.731	.729
gPb_NG [63]	.681	.716	.764	.687
Sketch Tokens [92]	.727	.746	.781	.727
MCG [131]	.749	.779	.759	.739
SCG [25]	.739	.758	.773	.747
SE [129]	.743	.763	.776	.743
OEF [65]	.749	.772	.817	.749
DeepNet [84]	.743	.764	.761	.771
PMI [73]	.742	.771	.797	.749
CSCNN [72]	.756	.775	.828	.756
LEP [191]	.757	.793	.797	.755
DeepContour [168]	.756	.766	.798	.757
DeepEdge [178]	.753	.772	.807	.753
DexiNed [130]	.728	.745	.689	.729
N4Fields [54]	.753	.769	.784	.754
Fined [170]	.788	.804	-	.787
CEDN [178]	.788	.804	-	.788
RDS [100]	.792	.810	.818	.792
COB [109]	.793	.820	.859	.793
RCF [99]	.798	.815	-	.806
HED [174]	.788	.808	.840	.788
PiDiNet [153]	.807	.823	-	.807
LPCB [34]	.800	.816	-	.808
RINDNet [134]	.800	.811	.815	.804
BDCN [66]	.806	.826	.847	.820
DSCD [33]	.822	.859	-	.822
RCN [82]	.823	.838	.853	.823
EDTER [135]	.824	.841	.880	.823

- Image edge detection is a fundamental task in the field of computer vision and image processing. However, its corresponding application has not been sufficiently explored. Thus, one of the promising research directions is to utilize modern image edge detection techniques for meeting different types of requirements of practical computer vision tasks such as image matching and 3D reconstruction.

6. Conclusion

In this work, the existing edge detection methods are reviewed in detail. Firstly, the definition and properties of edges are introduced. Secondly, the existing edge detection methods (i.e., hand-crafted based and machine learning based methods) are classified and introduced in detail. Thirdly, the existing widely used datasets and evaluation criteria for image edge detection methods are summarized.

Table 4

Comparison of evaluation results of state-of-the-art methods on the NYUD dataset [150]. In this experimental comparison, three different types of input representations are also considered, i.e., RGB, HHA [64], and RGB-HHA. The RGB-HHA is obtained by a weighted average over the model represented by RGB and HHA.

Method	ODS	OIS	AP	F-measure
gPb_ucm [8]	.632	.601	.562	.631
SE [129]	.651	.667	-	.651
SemiContour [190]	.680	.700	.690	.680
OEF [65]	.695	.708	.679	.695
RCF (RGB) [99]	.729	.742	-	.757
RCF (HHA)	.705	.715	-	.738
RCF (RGB_HHA)	.757	.771	-	.746
HED (RGB) [174]	.720	.734	.786	.741
HED (HHA)	.682	.695	.776	.726
HED (RGB_HHA)	.746	.761	.765	.734
LPCB (RGB) [34]	.739	.754	-	.762
LPCB (HHA)	.707	.719	-	.739
LPCB (RGB_HHA)	.762	.778	-	.748
BDCN (RGB) [66]	.748	.763	.770	.765
BDCN (HHA)	.707	.719	.762	.742
BDCN (RGB_HHA)	.765	.781	.748	.758
AMH (RGB) [175]	.744	.758	-	.759
AMH (HHA)	.716	.729	-	.744
AMH (RGB_HHA)	.771	.786	-	.750
PiDiNet (RGB) [153]	.733	.747	.765	.771
PiDiNet (HHA)	.715	.728	.734	.768
PiDiNet (RGB_HHA)	.756	.773	.802	.781
EDTER (RGB) [135]	.774	.789	.797	.780
EDTER (HHA)	.703	.718	.727	.778
EDTER (RGB_HHA)	.780	.797	.814	.782

Table 5

Comparison of evaluation results of state-of-the-art methods on the Multicue dataset [115].

Method	ODS	OIS	AP
HED [174]	.851(0.014)	.864(0.011)	-
RCF [99]	.857(0.004)	.862(0.004)	-
BDCN [66]	.891(0.001)	.898(0.002)	.935(0.002)
PiDiNet [153]	.855(0.007)	.860(0.005)	-
DSCD [33]	.871(0.007)	.876(0.002)	-
EDTER [135]	.894(0.005)	.900(0.003)	.944(0.002)

Finally, future research directions for image edge detection are elaborated. We expect that this survey will not only enable researchers to better understand image edge detection but also inspire future research activities.

Table 6

Comparison of evaluation results of state-of-the-art methods on the PASCAL dataset [48].

Method	ODS	OIS	AP	F-measure
HED [174]	.587	.598	.568	.587
COB [109]	.624	.657	.593	.624
CEDN [178]	.654	.657	.679	.654
RCN [82]	.752	.773	.641	.752



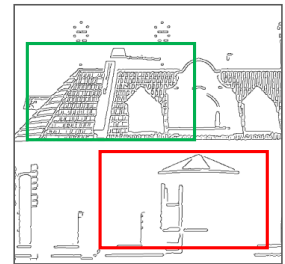
(a) In daylight



(b) Edge detection result

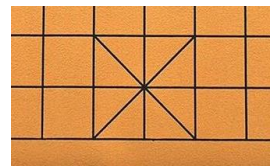


(c) In evening

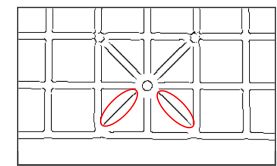


(d) Edge detection result

Figure 5: Examples of the edge detection results by an edge detector under different light conditions.



(a) Chessboard image



(b) Adjacent edge detection results

Figure 6: Examples of the adjacent edge detection results by an edge detector. (a) a local Chinese chessboard image, (b) the adjacent edge detection results from the HED method.

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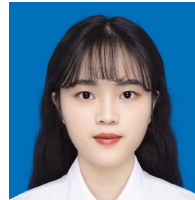
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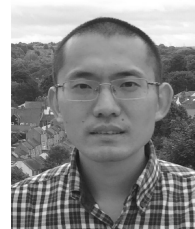
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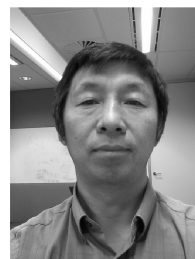
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